□ Data Structures (15 Marks)

✏ *Attempt all questions. Write algorithms where applicable.*

Short Answer Questions (1–2 Marks Each)

1. Define an array with one example.
2. What is a sparse matrix?
3. Define stack and mention any two applications.
4. What are the limitations of using arrays?

Medium Answer Questions (3–4 Marks Each)

5. Write the algorithm for **PUSH and POP** operations on a stack using array.
6. Convert the following **infix expression to postfix**:
   (A + B) * C – D
7. Explain the **array vs linked list representation** with example.
8. Write a short note on **multidimensional arrays**.

Long Answer (5 Marks)

9. Convert the following expression:
   Infix: (A + B) * (C – D)
   → Postfix and Prefix
   → Explain step-by-step
10. Define stack. Explain how to implement **multiple stacks** in a single array.

□ Section A: Very Short Answer (1 Mark each)

11. What is an array?
12. What is a stack?
13. Define a multidimensional array.
14. Expand LIFO.
15. Write the full form of SGPA. *(if asked in exam context)*
16. Define a sparse matrix.
17. What is the maximum size of a stack declared as int stack[100]?

□ Section B: Short Answer (2 Marks each)

18. Write the difference between 1D and 2D array.
19. List two applications of stack in real life.

20. Write any two limitations of arrays.
21. How is stack represented using arrays?
22. Define infix, prefix, and postfix expressions with examples.

□ Section C: Medium Answer (3–4 Marks each)

23. Convert the following infix expression to postfix:
    (A + B) * (C - D)
24. Explain how stack is implemented using arrays.
25. Represent the following sparse matrix using 3-tuple:

    0 0 5

    0 0 0

    3 0 0
26. Differentiate between array and linked list representation.
27. Write an algorithm for PUSH and POP operations in a stack.
28. Explain how multiple stacks can be implemented in a single array.

□ Section D: Long Answer (5 Marks each – Choose any one in exam)

29. Define a sparse matrix. Show how it is represented using a 3-tuple format.
30. Write a C-like program or pseudocode to implement stack operations using arrays.
31. Define stack and explain how it is used to convert infix expression to postfix with example.
32. Explain all steps to convert:
    Infix: A + (B * C - D)
    → **Prefix** and **Postfix**
33. Discuss applications of stack in expression evaluation and recursion.
34. What happens when stack overflows?
35. What is the role of 'top' in stack implementation?
36. Can a stack be implemented using a linked list? How?
37. Why is array not suitable for insertion and deletion in the middle?